



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2011 Preliminary Course Assessment Task 1

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 11M3A – Mr Weiss
- ☐ 11M3B – Mr Ireland
- ☐ 11M3C – Mr Berry
- ☐ 11M3D – Mrs Collins
- ☐ 11M3E – Mr Lam
- ☐ 11M3F – Mr Fletcher/Ms Everingham

NAME # SHEETS USED:

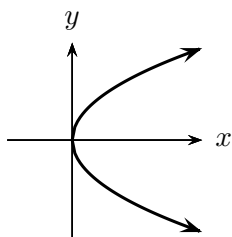
Marker's use only.

QUESTION	1	2	3	4	Total	%
MARKS	$\overline{17}$	$\overline{10}$	$\overline{31}$	$\overline{19}$	$\overline{77}$	

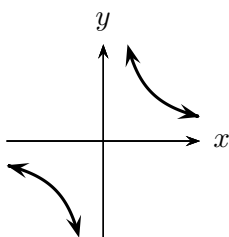
Question 1 (17 Marks) Commence a NEW page. **Marks**

(a) Which of the following are functions? **3**

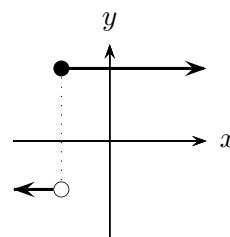
i.



ii.



iii.



(b) For the function $g(x) = \sqrt{x} + 3x$, find

i. $g(4)$ **1**

ii. $g(x^2)$ **1**

(c) Sketch, showing the essential features.

i. $y = |x| + 2$ **2**

ii. $y = \frac{4}{x-2}$ **2**

iii. $y = 4 - x^2$ **2**

iv. $(x-2)^2 + y^2 = 9$ **2**

(d) Consider the function $y = \sqrt{9 - x^2}$

i. State the domain and range. **2**

ii. Sketch the function. **1**

(e) Find the largest possible domain for $f(x) = \sqrt{x-4} + \sqrt{7-x}$. **1**

Question 2 (10 Marks) Commence a NEW page. **Marks**

(a) Express $0.\dot{3}\dot{7}\dot{8}$ as a simplest fraction. **2**

(b) If $2\sqrt{6} = \sqrt{x}$, find x . **1**

(c) Simplify $\sqrt{18} + \sqrt{8} - \sqrt{2}$ fully. **1**

(d) Expand and simplify:

i. $(2\sqrt{3} + \sqrt{5})^2$. **2**

ii. $(\sqrt{7} - 3)(\sqrt{7} + 3)$. **1**

(e) Rationalise the denominator and express in simplest form:

i. $\frac{1}{2\sqrt{3}}$. **1**

ii. $\frac{3}{2 + \sqrt{5}}$. **2**

Question 3 (31 Marks)

Commence a NEW page.

Marks

- (a) Solve for x :
- i. $x^2 = 2x$. **2**
 - ii. $3x^2 - 2x - 5 = 0$. **2**
 - iii. $2^x = \frac{1}{8}$. **2**
- (b) Expand and simplify:
- i. $(2m + n)(3m + 2n)$. **1**
 - ii. $3b + 2a - 7c - (2c + a + b)$. **1**
- (c) Simplify fully:
- i. $3a^2b \times 6a^3b^4$. **1**
 - ii. $\frac{30x^3y^2}{(6x^2)^3}$. **1**
- (d) Given $v^2 = u^2 - 2as$, find v correct to 3 significant figures if $u = 31.6$, $a = 9.8$ and $s = 40$. **2**
- (e) Factorise the following:
- i. $k^2 - 7k + 10$. **2**
 - ii. $27k^3 - 8t^3$. **2**
 - iii. $6d^2 + 5d - 21$. **2**
- (f) Simplify:
- i. $\frac{1}{a} + \frac{c}{b}$. **1**
 - ii. $\frac{2}{x} - \frac{3}{x(x+4)}$. **2**
 - iii. $\frac{(x^2 - 4)(3x + 1)}{(9x^2 - 1)(x + 2)}$. **2**
- (g) Solve simultaneously: $\begin{cases} y = 11x - 28 \\ y = x^2 - 4 \end{cases}$ **2**
- (h) Solve:
- i. $|3x + 2| < x - 3$ **3**
 - ii. $\frac{2}{x-1} \leq 3$ **3**

Question 4 (19 Marks)	Commence a NEW page.	Marks
(a) On separate sets of axes, sketch the functions $y = x $ and $y = \frac{x^2}{ x }$. Explain the difference between them.		3
(b) Is $f(x) = x^3 - x$ an even or odd function? Explain your answer.		2
(c) A function is defined as follows:		
$f(x) = \begin{cases} x + 2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x \leq 2 \\ -2 & \text{if } x > 2 \end{cases}$		
i. Find $f(-2) + f(3) + f(\sqrt{3})$.		1
ii. Sketch the graph of this function.		3
(d) Find the value of a for this continuous, piecewise defined function:		1
$f(x) = \begin{cases} 2 - x & \text{if } x < 2 \\ x^2 + a & \text{if } x \geq 2 \end{cases}$		
(e) Sketch the graph of $f(x) = \frac{x+4}{x-1}$, showing all important features.		4
(f) i. Sketch the graph of $y = x^2 - 6$, labelling all intercepts.		1
ii. On the same set of axes, carefully sketch the graph of $y = x $.		1
iii. Find the x coordinates of the two points where the graphs intersect.		2
iv. Hence solve the inequality $x^2 - 6 \leq x $.		1

End of paper.

Question 1

(a) (3 marks) – (ii) & (iii) are functions.

(b) i. (1 mark)

$$g(4) = \sqrt{4} + 3 \times 4 = 14$$

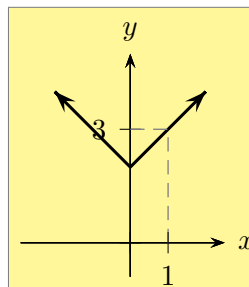
ii. (1 mark)

$$g(x^2) = \sqrt{x^2} + 3x^2 = x + 3x^2$$

(c) i. (2 marks)

✓ [1] for shape

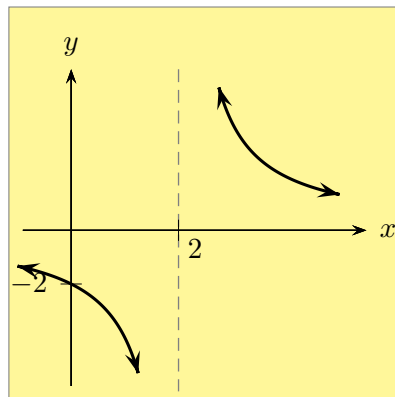
✓ [1] for showing 2 points.



ii. (2 marks)

✓ [1] for shape

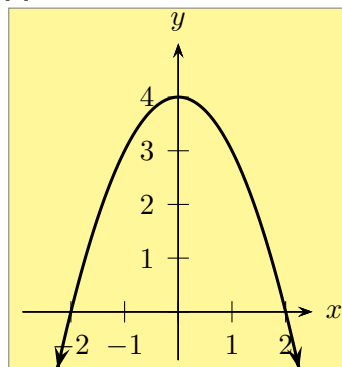
✓ [1] for asymptotes.



iii. (2 marks)

✓ [1] for shape

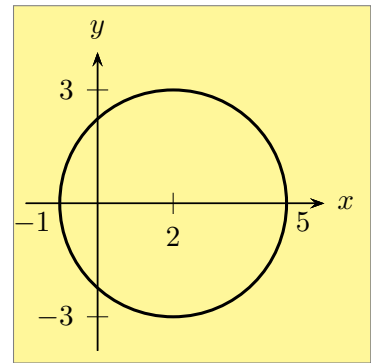
✓ [1] for intercepts.



iv. (2 marks)

✓ [1] for shape.

✓ [1] for centre.

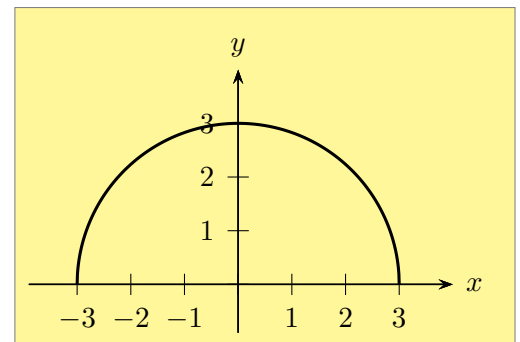


(d) i. (2 marks)

$$D = \{x : -3 \leq x \leq 3\}$$

$$R = \{y : 0 \leq y \leq 3\}$$

ii. (1 mark)



(e) (1 mark)

$$f(x) = \sqrt{x-4} + \sqrt{7-x}$$

Domain for $y = \sqrt{x-4}$:

$$x - 4 \geq 0$$

$$\therefore x \geq 4$$

Domain for $y = \sqrt{7-x}$:

$$y = \sqrt{7-x} = \sqrt{-(x-7)}$$

$$-(x-7) \geq 0$$

$$x - 7 \leq 0$$

$$x \leq 7$$

Hence the domain of $y = \sqrt{x-4} + \sqrt{7-x}$ is

$$D = \{x : 4 \leq x \leq 7\}$$

Question 2

(a) (2 marks)

- ✓ [1] for $1000x$.
 ✓ [1] for final answer.

$$\begin{aligned}x &= 0.378\overline{78} \\10x &= 3.7878\overline{78} \\1000x &= 378.7878\overline{78} \\\therefore 990x &= 375\end{aligned}$$

$$x = \frac{375}{990} = \frac{25}{66}$$

(b) (1 mark)

$$\begin{aligned}2\sqrt{6} &= \sqrt{24} \\\therefore x &= 24\end{aligned}$$

(c) (1 mark)

$$\begin{aligned}\sqrt{18} + \sqrt{8} - \sqrt{2} &= 3\sqrt{2} + 2\sqrt{2} - \sqrt{2} \\&= 4\sqrt{2}\end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for expanding correctly.
 ✓ [1] for simplifying expanded answer.

$$\begin{aligned}(2\sqrt{3} + \sqrt{5})^2 &= 12 + 5 + 4\sqrt{15} \\&= 17 + 4\sqrt{15}\end{aligned}$$

ii. (1 mark)

$$(\sqrt{7} - 3)(\sqrt{7} + 3) = 7 - 9 = -2 \quad \text{(b) i. (1 mark)}$$

(e) i. (1 mark)

$$\frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{6}$$

ii. (2 marks)

- ✓ [1] for multiplying by conjugate of denominator in numerator & denominator.
 ✓ [1] for final answer.

$$\begin{aligned}\frac{3}{2 + \sqrt{5}} \times \frac{2 - \sqrt{5}}{2 - \sqrt{5}} &= \frac{6 - 3\sqrt{5}}{4 - 5} \\&= 3\sqrt{5} - 6\end{aligned}$$

Question 3

(a) i. (2 marks)

- ✓ [1] for changing subject to "0".
 ✓ [1] for final solutions.

$$\begin{aligned}x^2 &= 2x \\x^2 - 2x &= 0 \\x(x - 2) &= 0 \\\therefore x &= 0, 2\end{aligned}$$

ii. (2 marks)

- ✓ [1] for correct factorisation.
 ✓ [1] for final solutions.

$$\begin{aligned}3x^2 - 2x - 5 &= 0 \\(3x - 5)(x + 1) &= 0 \\\therefore x &= \frac{5}{3}, -1\end{aligned}$$

iii. (2 marks)

- ✓ [1] for identifying $\frac{1}{8} = 2^{-3}$.
 ✓ [1] for final solution.

$$\begin{aligned}2^x &= \frac{1}{8} = 2^{-3} \\\therefore x &= -3\end{aligned}$$

$$(2m + n)(3m + 2n) = 6m^2 + 7mn + 2n^2$$

ii. (1 mark)

$$\begin{aligned}3b + 2a - 7c - (2c + a + b) \\&= 3b + 2a - 7c - 2c - a - b \\&= a + 2b - 9c\end{aligned}$$

- (c) i. (1 mark)

$$3a^2b \times 6a^3b^4 = 18a^5b^5$$

- ii. (1 mark)

$$\frac{30x^3y^2}{(6x^2)^3} = \frac{\cancel{3} \times 5 \times \cancel{2}^2 y^2}{6^2 \times \cancel{6} \times \cancel{x^6}_{x^3}} = \frac{5y^2}{36x^3}$$

- (d) (2 marks)

✓ [1] for $v^2 = 214.56$.✓ [1] for v to 3 s.f.

- iii. (2 marks)

✓ [1] for factorising fully.

✓ [1] for final answer.

$$\begin{aligned} & \frac{(x^2 - 4)(3x + 1)}{(9x^2 - 1)(x + 2)} \\ &= \frac{\cancel{(x+2)}(x-2)\cancel{(3x+1)}}{(3x-1)\cancel{(3x+1)}\cancel{(x+2)}} \\ &= \frac{x-2}{3x-1} \end{aligned}$$

- (g) (2 marks)

$$\begin{aligned} v^2 &= u^2 - 2as \\ v^2 &= (31.6)^2 - 2 \times 9.8 \times 40 \\ &= 214.56 \\ \therefore v &= 14.6 \text{ (3 s.f.)} \end{aligned}$$

$$\begin{cases} y = 11x - 28 \\ y = x^2 - 4 \end{cases}$$

Equating,

$$\begin{aligned} x^2 - 4 &= 11x - 28 \\ x^2 - 11x + 24 &= 0 \\ (x - 3)(x - 8) &= 0 \\ \therefore x &= 3, 8 \\ \therefore y &= 5, 60 \end{aligned}$$

Hence the point of intersection is (3, 5) and (8, 60).

- (e) i. (2 marks)

$$k^2 - 7k + 10 = (k - 5)(k - 2)$$

- ii. (2 marks)

$$27k^3 - 8t^3 = (3k - 2t)(9k^2 + 6kt + 4t^2)$$

- iii. (2 marks)

$$6d^2 + 5d - 21 = (3d + 7)(2d - 3)$$

(h)

- i. (3 marks)

✓ [2] for finding x intercepts $-\frac{5}{2}$ and $\frac{1}{4}$.

✓ [1] for final solution.

- (f) i. (1 mark)

$$\frac{1}{a} + \frac{c}{b} = \frac{b + ac}{ab}$$

- ii. (2 marks)

✓ [1] for creating a common denominator.

✓ [1] for final answer.

$$\begin{aligned} \frac{2}{x} - \frac{3}{x(x+4)} &= \frac{2(x+4)}{x(x+4)} - \frac{3}{x(x+4)} \\ &= \frac{2x+8-3}{x(x+4)} \\ &= \frac{2x+5}{x(x+4)} \end{aligned}$$

$$|3x + 2| < x - 3$$

Squaring both sides,

$$\begin{aligned} (3x + 2)^2 &< (x - 3)^2 \\ (3x + 2)^2 - (x - 3)^2 &< 0 \\ [(3x + 2) - (x - 3)][(3x + 2) + (x - 3)] &< 0 \\ (2x + 5)(4x - 1) &< 0 \\ \therefore -\frac{5}{2} &< x < \frac{1}{4} \end{aligned}$$

Testing for $x = 0$ (which falls within the interval $-\frac{5}{2} < x < \frac{1}{4}$):

$$\begin{aligned} |3(0) + 2| &= 2 \\ 0 - 3 &= -3 \end{aligned}$$

but $2 \not< -3$. Hence there are no real solutions.

Alternatively, examine cases:

- $3x + 2 \geq 0$, i.e. $x \geq -\frac{2}{3}$

$$3x + 2 < x - 3$$

$$2x < -5$$

$$x < -\frac{5}{2}$$

But there is no solution such that $x \geq -\frac{2}{3}$ and $x < -\frac{5}{2}$. Hence

no solution to this branch.

- $3x + 2 < 0$, i.e. $x < -\frac{2}{3}$.

$$-(3x + 2) < x - 3$$

$$-3x - 2 < x - 3$$

$$4x > 1$$

$$x > \frac{1}{4}$$

Again there is no x that exists such that $x > \frac{1}{4}$ and $x < -\frac{2}{3}$. So

no solution to this branch.

Hence there are no real solutions.

ii. (3 marks)

- ✓ [1] for multiplying by square of denominator on both sides.
- ✓ [1] for obtaining $(x-1)(3x-5) \geq 0$.
- ✓ [1] for final solution.

$$\frac{2}{x-1} \leq \frac{3}{x(x-1)^2}$$

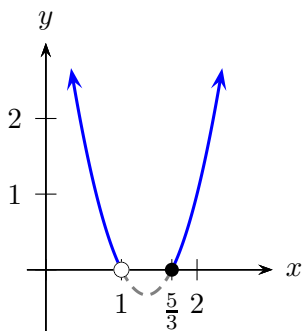
Note: $x \neq 1$

$$2(x-1) \leq 3(x-1)^2$$

$$3(x-1)^2 - 2(x-1) \geq 0$$

$$(x-1)(3(x-1)-2) \geq 0$$

$$(x-1)(3x-5) \geq 0$$



As $x \neq 1$,

$$x < 1 \text{ or } x \geq \frac{5}{3}$$

Alternatively,

$$\begin{aligned} \frac{2}{x-1} &\leq 3 \\ \frac{2}{x-1} - \frac{3(x-1)}{x-1} &\leq 0 \\ \frac{2-3x+3}{x-1} &\leq 0 \\ \frac{-3x+5}{x-1} &\leq 0 \end{aligned}$$

Drawing a table of values for $-3x+5$ and $x-1$ and testing values for each of these expressions (values tested are shown in parentheses):

x	(0)	1	(1.1)	$\frac{5}{3}$	(2)
sign of: $-3x+5$	+	+	0	-	-
sign of: $x-1$	-	-	+	+	+
result: $\frac{-3x+5}{x-1}$	-	-	+	0	-

From the table,

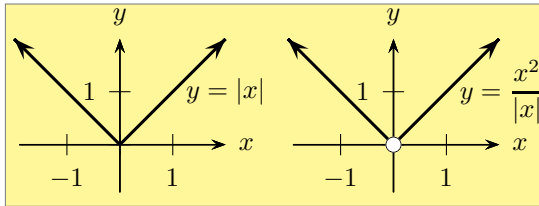
$$\frac{-3x+5}{x-1} \leq 0$$

when $x < 1$ or $x \geq \frac{5}{3}$.

Question 4

(d) (1 mark)

(a) (3 marks)



$$f(x) = \begin{cases} 2 - x & \text{if } x < 2 \\ x^2 + a & \text{if } x \geq 2 \end{cases}$$

$$f(2^-) = 2 - 2 = 0$$

$$\therefore f(2) = 2^2 + a = 0$$

$$\therefore a = -4$$

The graph of $y = \frac{x^2}{|x|}$ is essentially the same as $y = |x|$ but $x \neq 0$ for the former. (e) (4 marks)

(b) (2 marks)

✓ [2] for hyperbola.

✓ [2] for asymptotes.

$$f(x) = x^3 - x$$

$$f(-x) = (-x)^3 - (-x) \quad \left| \quad -f(x) = -(x^3 - x) \right.$$

$$= -x^3 + x \quad \left| \quad = -x^3 + x \right.$$

As $f(-x) = -f(x)$, the function is odd.

$$y = \frac{x+4}{x-1} = \frac{x-1+5}{x-1}$$

$$= \frac{x-1}{x-1} + \frac{5}{x-1} = 1 + \frac{5}{x-1}$$

(c) i. (1 mark)

$$f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \leq x \leq 2 \\ -2 & \text{if } x > 2 \end{cases}$$

$$f(-2) = -2 + 2 = 0$$

$$f(3) = -2$$

$$f(\sqrt{3}) = (\sqrt{3})^2 = 3$$

$$\therefore f(-2) + f(3) + f(\sqrt{3}) = 0 - 2 + 3$$

$$= 1$$

Confirming the asymptotes:

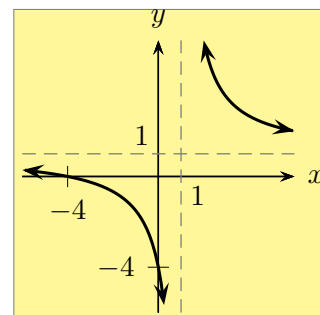
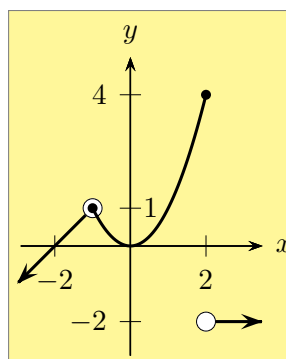
$$y = \frac{5}{x-1} + 1$$

$$(y-1)(x-1) = 5$$

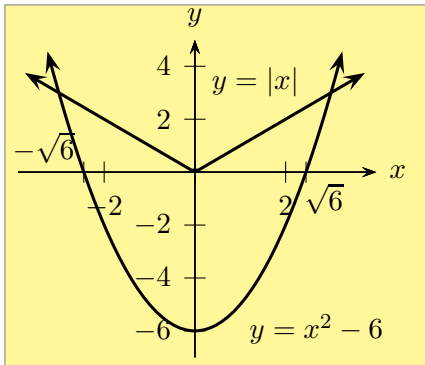
$$\therefore y \neq 1 \quad x \neq 1$$

ii. (3 marks)

✓ [1] for each piece of the graph.



(f) i. (1 mark)



ii. (1 mark) – see above.

iii. (2 marks)

$$x^2 - 6 = |x|$$

Case 1: $x \geq 0$

$$x^2 - 6 = x$$

$$x^2 - x - 6 = 0$$

$$(x - 3)(x + 2) = 0$$

$$\therefore x = 3 \text{ or } -2$$

Testing values:

- $x = -2$:

$$|-2| = 2$$

$$(-2)^2 - 6 \neq 2 \quad \times$$

- $x = 3$:

$$|3| = 3$$

$$3^2 - 6 = 9 - 6 = 3 \quad \checkmark$$

Case 2: $x < 0$:

$$x^2 - 6 = -x$$

$$x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0$$

$$x = -3, 2$$

Testing values:

- $x = -3$:

$$|-3| = 3$$

$$(-3)^2 - 6 = 3 \quad \checkmark$$

- $x = 2$:

$$|2| = 2$$

$$2^2 - 6 = 4 - 6 = -2 \quad \times$$

$$\therefore x = -3, 3$$

Note it is possible to find the points of intersection graphically.

iv. (1 mark)

$$-3 \leq x \leq 3$$